

# Beyond Scaling: Measuring and Predicting the Upper Bound of Knowledge Retention in Language Model Pre-Training

Anonymous submission

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## B Templates for Knowledge Triples

We create twenty templates for each type of knowledge triple. Table 1 illustrates the templates for the BORN relation, along with their corresponding evaluation accuracy on the ours-13b model. As shown, the choice of template has a significant impact on the model’s accuracy in answering questions. To address the instability of evaluations based on a single template, we calculate the average accuracy across all 20 templates. We employ **gpt-4o-2024-08-06** to generate query templates. The prompt we use is: **Generate 20 diverse and fluent paraphrases of the following sentence, preserving its original meaning:\n[sentence]**. Our evaluation set contains a total of 12,671 pieces of data and 300 query templates, following the MIT license.

Table 1: Twenty templates and their corresponding accuracy values for the BORN relation on the ours-13b model. [S] represents the subject in the knowledge triple.

| Template                                     | ACC   |
|----------------------------------------------|-------|
| [S]’s birth took place in                    | 0.186 |
| [S] was brought into existence in            | 0.187 |
| [S] was born into the world in               | 0.190 |
| [S]’s nativity is                            | 0.197 |
| The place in which [S] was given birth to is | 0.201 |
| The hometown of [S] is                       | 0.203 |
| The birthplace of [S] is                     | 0.212 |
| [S] was delivered in                         | 0.215 |
| [S]’s cradle was in                          | 0.219 |
| [S]’s entry into the world happened in       | 0.225 |
| [S] first saw the light in                   | 0.228 |
| [S]’s roots are in                           | 0.230 |
| [S]’s life began in                          | 0.230 |
| [S] was born and raised in                   | 0.237 |
| [S]’s origin is                              | 0.238 |
| [S] entered life in                          | 0.251 |
| [S] came into the world in                   | 0.255 |
| [S] was born in                              | 0.255 |
| [S]’s birth occurred in                      | 0.263 |
| [S]’s starting point was                     | 0.282 |

## C Case Study

Table 2 highlights four counterintuitive cases in our pre-training data. From these, we observe the following: (1) In the first two cases, the subject and object rarely co-occur, suggesting a seemingly weak relationship. However, whenever the subject does appear, the object almost always appears alongside it, resulting in high accuracy. (2) In the last two cases, despite frequent co-occurrence between the subject and object, their individual occurrence frequencies are significantly higher, leading to low accuracy. For example, when prompted with “Paris is located in”, the model is more likely to respond with “France” rather than “Europe”.

Table 2: Several cases with significant deviations from the co-occurrence metric are shown here.  $N_s$  is the number of paragraphs in which the subject appears,  $N_o$  is the number of paragraphs in which the object appears, and  $N_{s,o}$  is the number of paragraphs in which they co-occur. The first two cases exhibit rare co-occurrence but demonstrate high accuracy, whereas the latter two cases show frequent co-occurrence but exhibit low accuracy.

| Subject                                 | Question                            | Object   | $N_s$      | $N_o$      | $N_{s,o}$ | ACC   |
|-----------------------------------------|-------------------------------------|----------|------------|------------|-----------|-------|
| Rare Co-occurrence but High Accuracy    |                                     |          |            |            |           |       |
| Csepel SC                               | The headquarters of Csepel SC is in | Budapest | 398        | 748,860    | 172       | 0.655 |
| Liis Lemsalu                            | Liis Lemsalu is a citizen of        | Estonia  | 307        | 967,207    | 160       | 0.570 |
| Frequent Co-occurrence but Low Accuracy |                                     |          |            |            |           |       |
| Paris                                   | Paris is located in                 | Europe   | 20,902,571 | 59,408,155 | 3,761,610 | 0.043 |
| Ohio                                    | The capital of Ohio is              | Columbus | 11,894,303 | 3,718,483  | 1,210,957 | 0.193 |

These examples highlight the pivotal role of knowledge specificity in the memory of LLMs. It is not simply the frequencies of knowledge occurrence that enhance retention, but rather the specificity of the knowledge that proves more influential. This insight suggests that in downstream task training, even when domain-specific data is significantly fewer than the pre-training data, minimizing repetition with the pre-training data can increase the SMI, making it easier for LLMs to retain the relevant knowledge.

## References

- Biderman, S.; Schoelkopf, H.; Anthony, Q. G.; Bradley, H.; O’Brien, K.; Hallahan, E.; Khan, M. A.; Purohit, S.; Prashanth, U. S.; Raff, E.; Skowron, A.; Sutawika, L.; and van der Wal, O. 2023. Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling. In Krause, A.; Brunskill, E.; Cho, K.; Engelhardt, B.; Sabato, S.; and Scarlett, J., eds., *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA*, volume 202 of *Proceedings of Machine Learning Research*, 2397–2430. PMLR.
- Black, S.; Biderman, S.; Hallahan, E.; Anthony, Q.; Gao, L.; Golding, L.; He, H.; Leahy, C.; McDonnell, K.; Phang, J.; Pieler, M.; Prashanth, U. S.; Purohit, S.; Reynolds, L.; Tow, J.; Wang, B.; and Weinbach, S. 2022. GPT-NeoX-20B: An Open-Source Autoregressive Language Model. *CoRR*, abs/2204.06745.
- Black, S.; Leo, G.; Wang, P.; Leahy, C.; and Biderman, S. 2021. GPT-Neo: Large Scale Autoregressive Language Modeling with Mesh-Tensorflow. If you use this software, please cite it using these metadata.
- Elazar, Y.; Kassner, N.; Ravfogel, S.; Ravichander, A.; Hovy, E. H.; Schütze, H.; and Goldberg, Y. 2021. Measuring and Improving Consistency in Pretrained Language Models. *Trans. Assoc. Comput. Linguistics*, 9: 1012–1031.
- Foundation, W. ????. Wikimedia Downloads.
- Gao, L.; Biderman, S.; Black, S.; Golding, L.; Hoppe, T.; Foster, C.; Phang, J.; He, H.; Thite, A.; Nabeshima, N.; Presser, S.; and Leahy, C. 2021. The Pile: An 800GB Dataset of Diverse Text for Language Modeling. *CoRR*, abs/2101.00027.
- Laurençon, H.; Saulnier, L.; Wang, T.; Akiki, C.; del Moral, A. V.; Scao, T. L.; von Werra, L.; Mou, C.; Ponferrada, E. G.; Nguyen, H.; Frohberg, J.; Sasko, M.; Lhoest, Q.; McMillan-Major, A.; Dupont, G.; Biderman, S.; Rogers, A.; Allal, L. B.; Toni, F. D.; Pistilli, G.; Nguyen, O.; Nikpoor, S.; Masoud, M.; Colombo, P.; de la Rosa, J.; Villegas, P.; Thrush, T.; Longpre, S.; Nagel, S.; Weber, L.; Muñoz, M.; Zhu, J.; van Strien, D.; Alyafeai, Z.; Almubarak, K.; Vu, M. C.; Gonzalez-Dios, I.; Soroa, A.; Lo, K.; Dey, M.; Suarez, P. O.; Gokaslan, A.; Bose, S.; Adelani, D. I.; Phan, L.; Tran, H.; Yu, I.; Pai, S.; Chim, J.; Lepercq, V.; Ilic, S.; Mitchell, M.; Luccioni, A. S.; and Jernite, Y. 2022. The BigScience ROOTS Corpus: A 1.6TB Composite Multilingual Dataset. In Koyejo, S.; Mohamed, S.; Agarwal, A.; Belgrave, D.; Cho, K.; and Oh, A., eds., *Advances in Neural Information*

*Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022.*

Penedo, G.; Malartic, Q.; Hesslow, D.; Cojocaru, R.; Alobeidli, H.; Cappelli, A.; Pannier, B.; Almazrouei, E.; and Launay, J. 2023. The RefinedWeb Dataset for Falcon LLM: Outperforming Curated Corpora with Web Data Only. In Oh, A.; Naumann, T.; Globerson, A.; Saenko, K.; Hardt, M.; and Levine, S., eds., *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023.*

Scao, T. L.; Fan, A.; Akiki, C.; Pavlick, E.; Ilic, S.; Hesslow, D.; Castagné, R.; Luccioni, A. S.; Yvon, F.; Gallé, M.; Tow, J.; Rush, A. M.; Biderman, S.; Webson, A.; Ammanamanchi, P. S.; Wang, T.; Sagot, B.; Muennighoff, N.; del Moral, A. V.; Ruwase, O.; Bawden, R.; Bekman, S.; McMillan-Major, A.; Beltagy, I.; Nguyen, H.; Saulnier, L.; Tan, S.; Suarez, P. O.; Sanh, V.; Laurençon, H.; Jernite, Y.; Launay, J.; Mitchell, M.; Raffel, C.; Gokaslan, A.; Simhi, A.; Soroa, A.; Aji, A. F.; Alfassy, A.; Rogers, A.; Nitzav, A. K.; Xu, C.; Mou, C.; Emezue, C.; Klamm, C.; Leong, C.; van Strien, D.; Adelani, D. I.; and et al. 2022. BLOOM: A 176B-Parameter Open-Access Multilingual Language Model. *CoRR*, abs/2211.05100.

Soboleva, D.; Al-Khateeb, F.; Myers, R.; Steeves, J. R.; Hestness, J.; and Dey, N. 2023. SlimPajama: A 627B token cleaned and deduplicated version of RedPajama. <https://www.cerebras.net/blog/slimpajama-a-627b-token-cleaned-and-deduplicated-version-of-redpajama>.

Wang, B. 2021. Mesh-Transformer-JAX: Model-Parallel Implementation of Transformer Language Model with JAX. <https://github.com/kingoflolz/mesh-transformer-jax>.

Zhang, P.; Zeng, G.; Wang, T.; and Lu, W. 2024. TinyLlama: An Open-Source Small Language Model. *CoRR*, abs/2401.02385.